

Chapter 5

Network Layer: Control Plane

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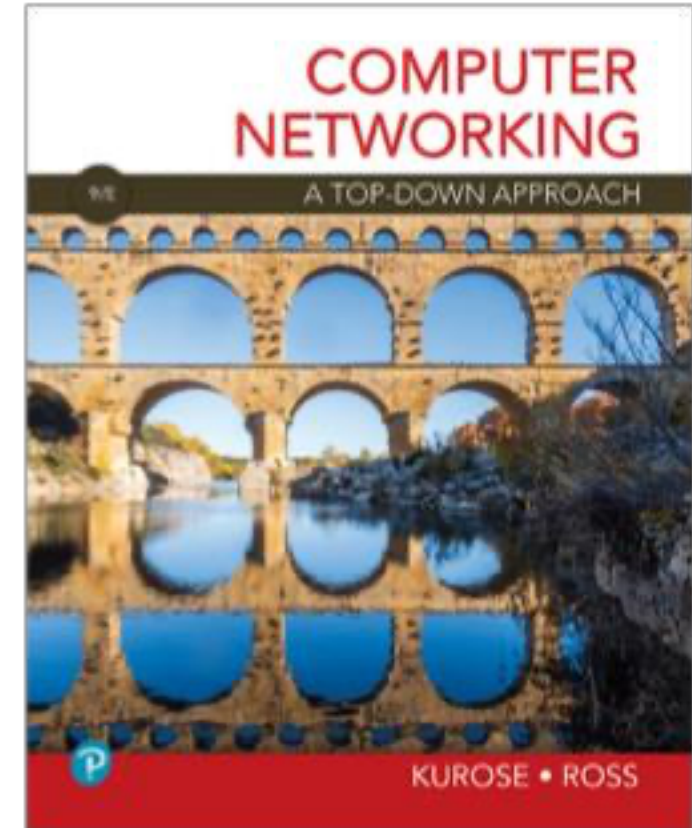
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*Computer Networking: A
Top-Down Approach*

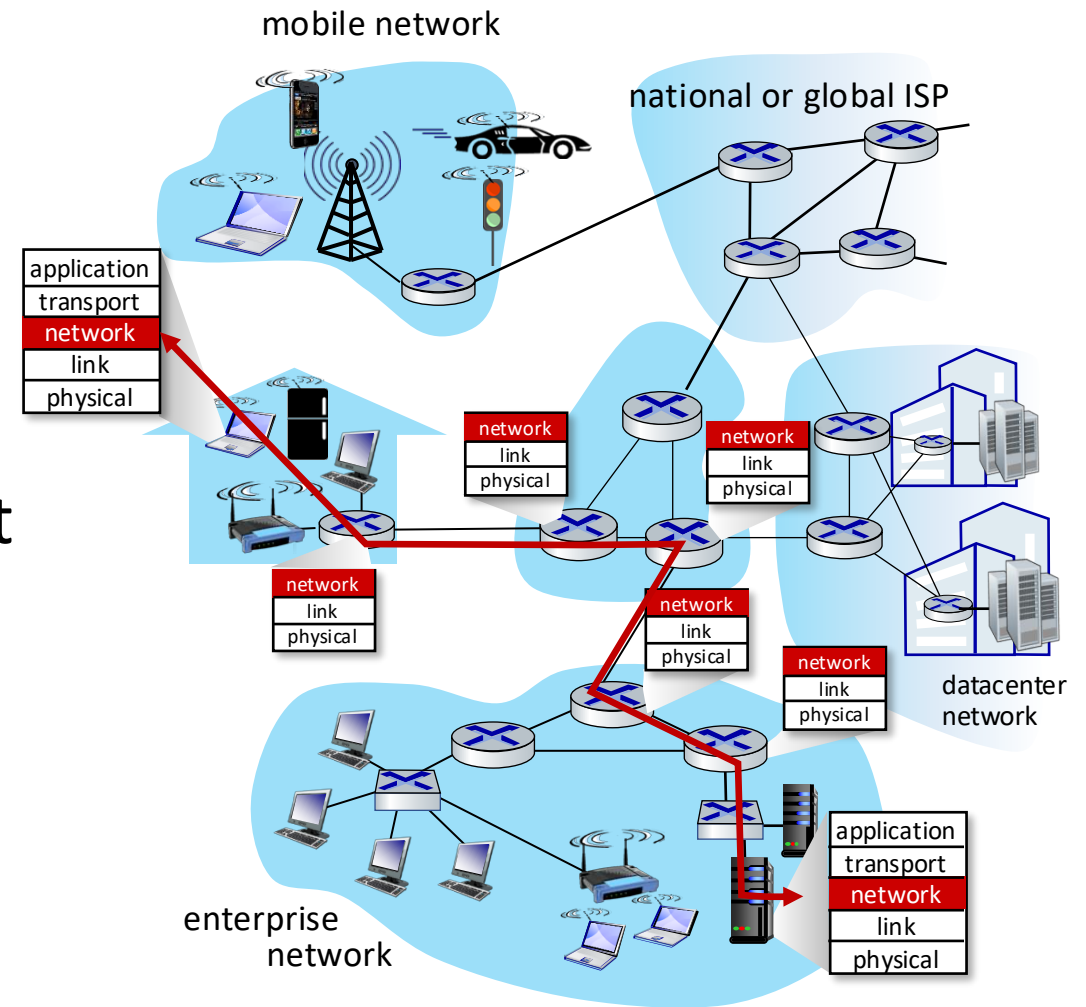
9th edition

Jim Kurose, Keith Ross
Pearson, 2025

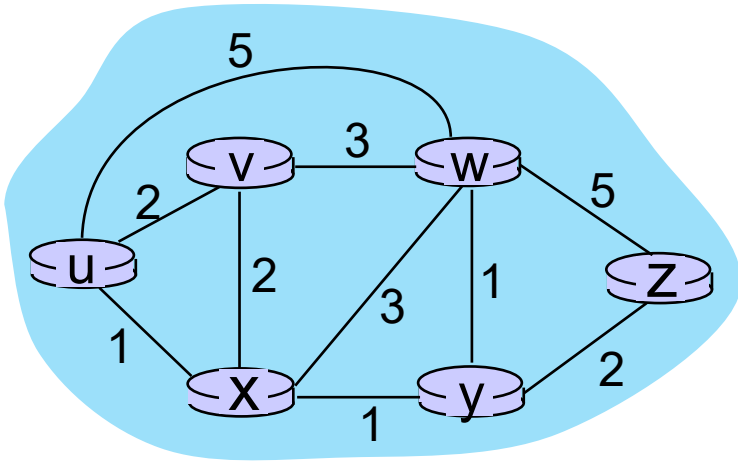
Routing protocols

Routing protocol goal: determine “good” paths (equivalently, routes), from sending hosts to receiving host, through network of routers

- **path:** sequence of routers packets traverse from given initial source host to final destination host
- **“good”:** least “cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!



Graph abstraction: link costs



$c_{a,b}$: cost of *direct* link connecting a and b

e.g., $c_{w,z} = 5$, $c_{u,z} = \infty$

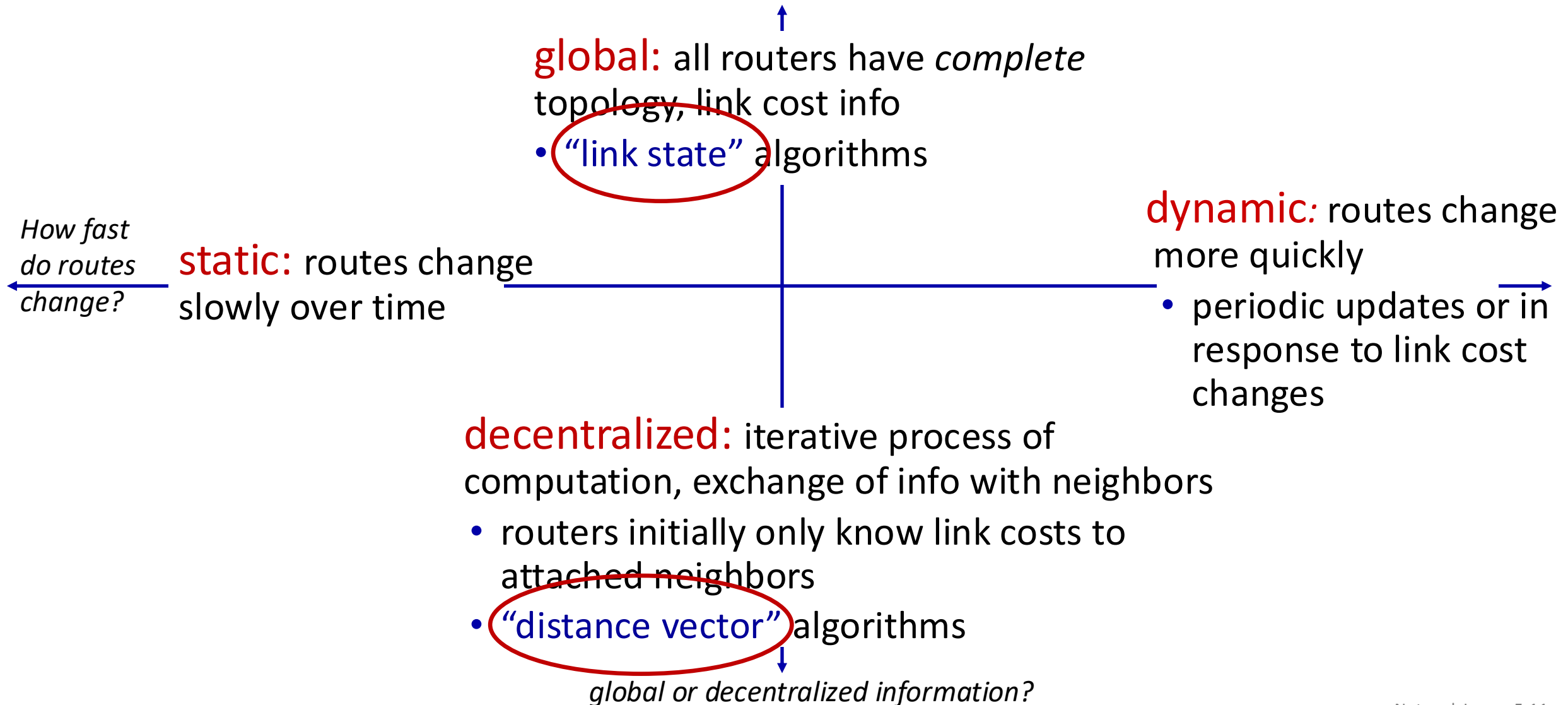
cost defined by network operator:
could always be 1, or inversely related
to bandwidth, or inversely related to
congestion

graph: $G = (N, E)$

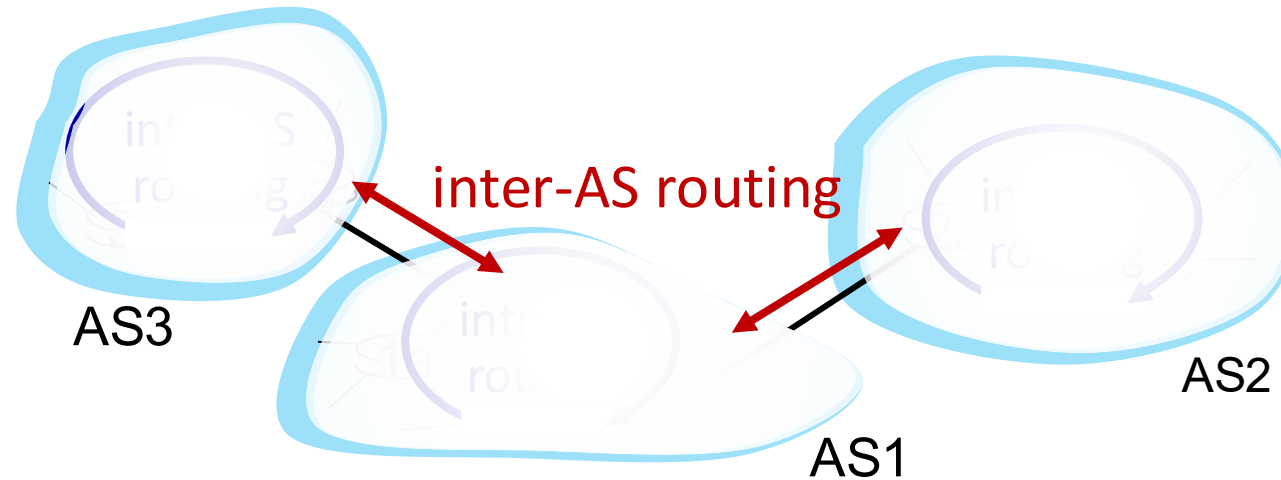
N : set of routers = $\{ u, v, w, x, y, z \}$

E : set of links = $\{ (u,v), (u,x), (v,x), (v,w), (x,w), (x,y), (w,y), (w,z), (y,z) \}$

Routing algorithm classification



Interconnected ASes



- ✓ **intra-AS (aka “intra-domain”)**: routing among routers *within same* AS (“*network*”)
- ➡ **inter-AS (aka “inter-domain”)**: routing *among* AS'es

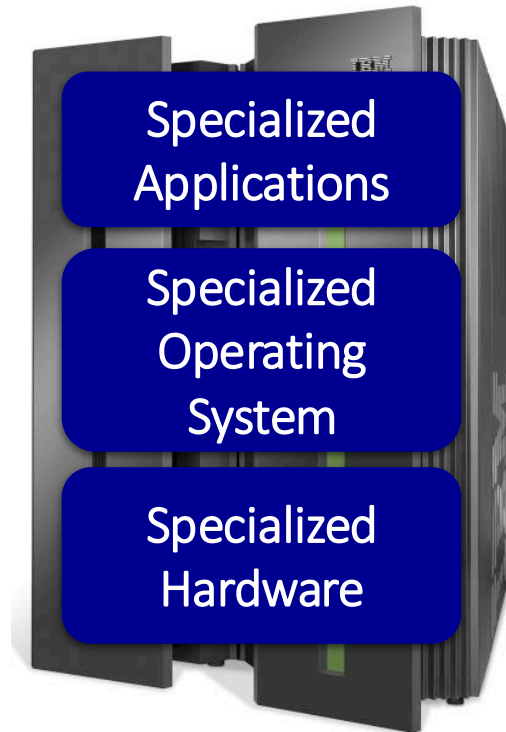
Internet inter-AS routing: BGP

- **BGP (Border Gateway Protocol)**: *the* de facto inter-domain routing protocol
 - “glue that holds the Internet together”
- allows subnet to advertise its existence, and the destinations it can reach, to rest of Internet: *“I am here, here is who I can reach, and how”*
- BGP provides each AS a means to:
 - obtain destination network reachability info from neighboring ASes (**eBGP**)
 - determine routes to other networks based on reachability information and *policy*
 - propagate reachability information to all AS-internal routers (**iBGP**)
 - **advertise** (to neighboring networks) destination reachability info

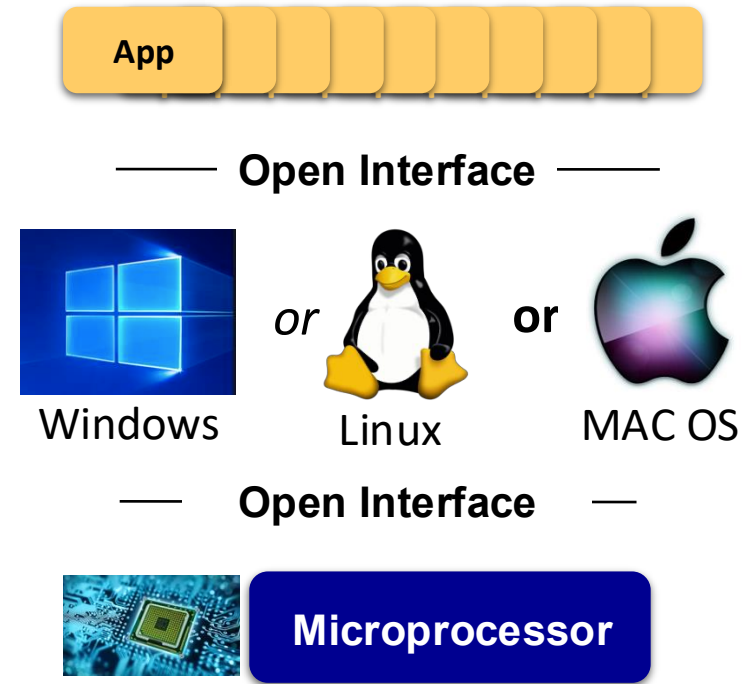
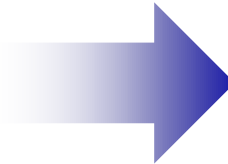
Software defined networking (SDN)

- Internet network layer: historically implemented via distributed, per-router control approach:
 - *monolithic* router contains switching hardware, runs proprietary implementation of Internet standard protocols (IP, RIP, IS-IS, OSPF, BGP) in proprietary router OS (e.g., Cisco IOS)
 - different “middleboxes” for different network layer functions: firewalls, load balancers, NAT boxes, ..
- ~2005: renewed interest in rethinking network control plane

SDN analogy: mainframe to PC revolution

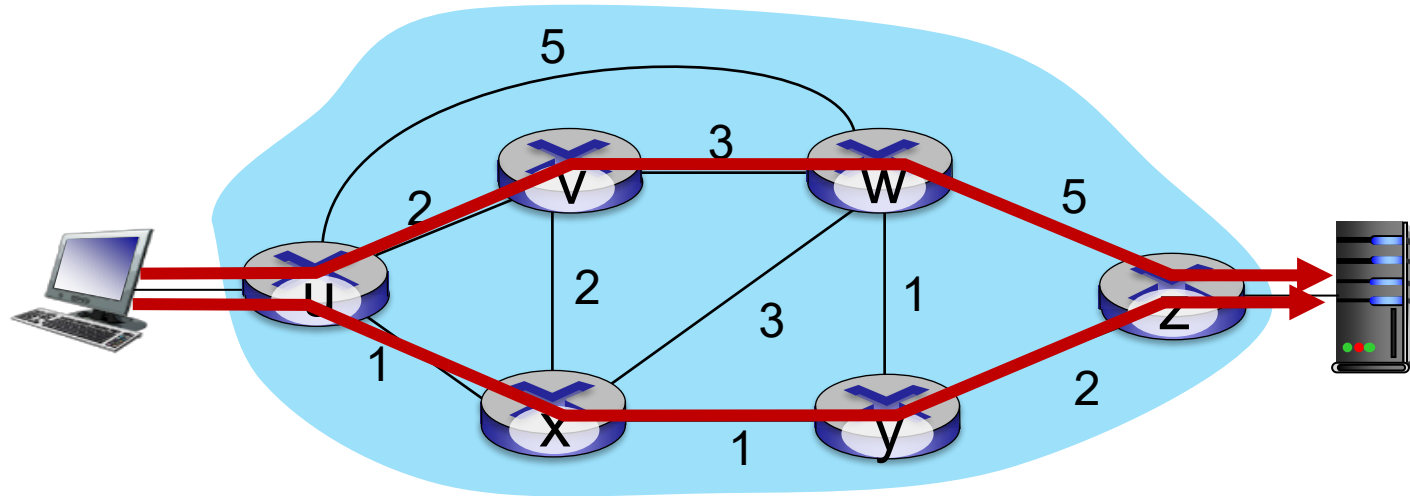


Vertically integrated
Closed, proprietary
Slow innovation
Small industry



Horizontal
Open interfaces
Rapid innovation
Huge industry

Traffic engineering: difficult with traditional routing

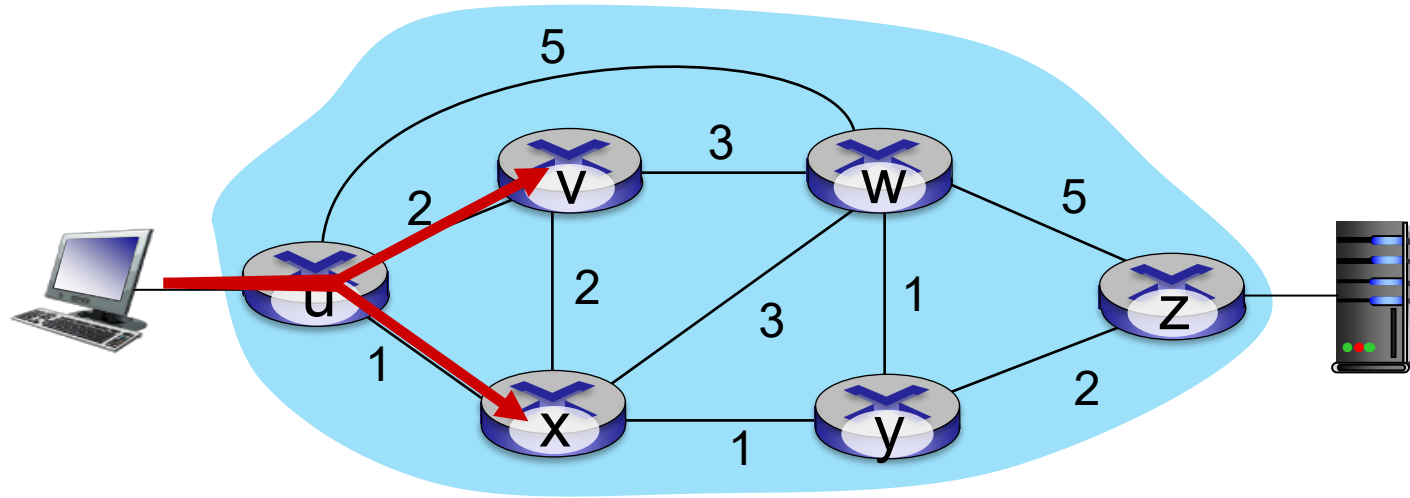


Q: what if network operator wants u-to-z traffic to flow along *uvwz*, rather than *uxyz*?

A: need to re-define link weights so traffic routing algorithm computes routes accordingly (or need a new routing algorithm)!

link weights are only control “knobs”: not much control!

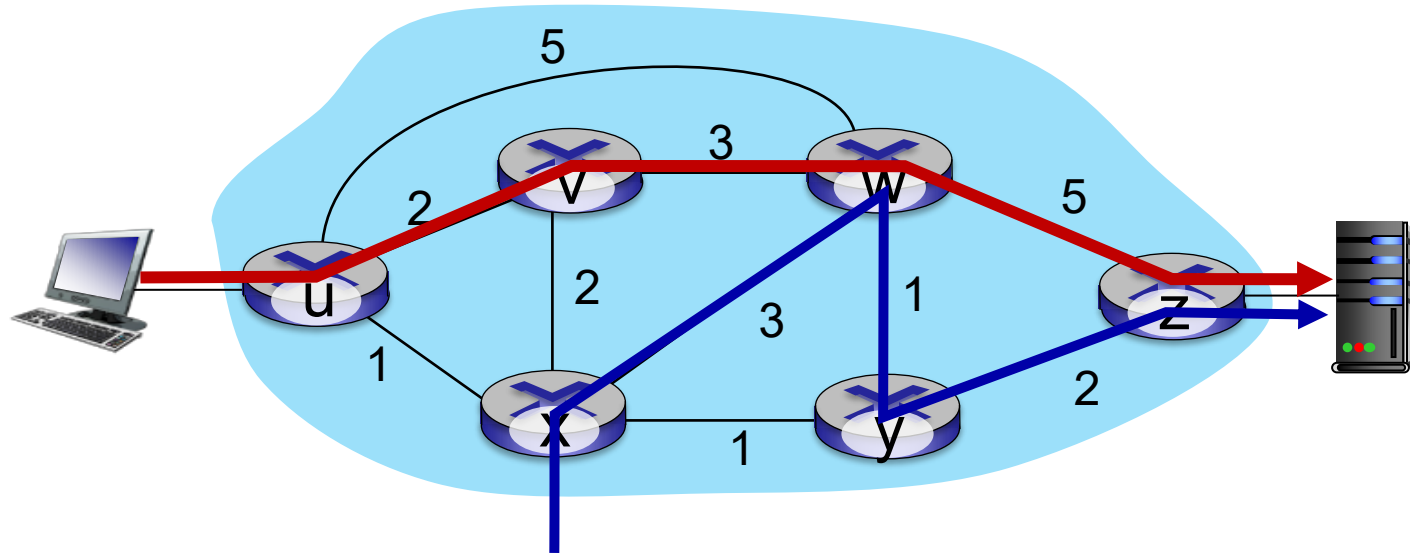
Traffic engineering: difficult with traditional routing



Q: what if network operator wants to split u-to-z traffic along uvwz *and* uxyz (load balancing)?

A: can't do it (or need a new routing algorithm)

Traffic engineering: difficult with traditional routing

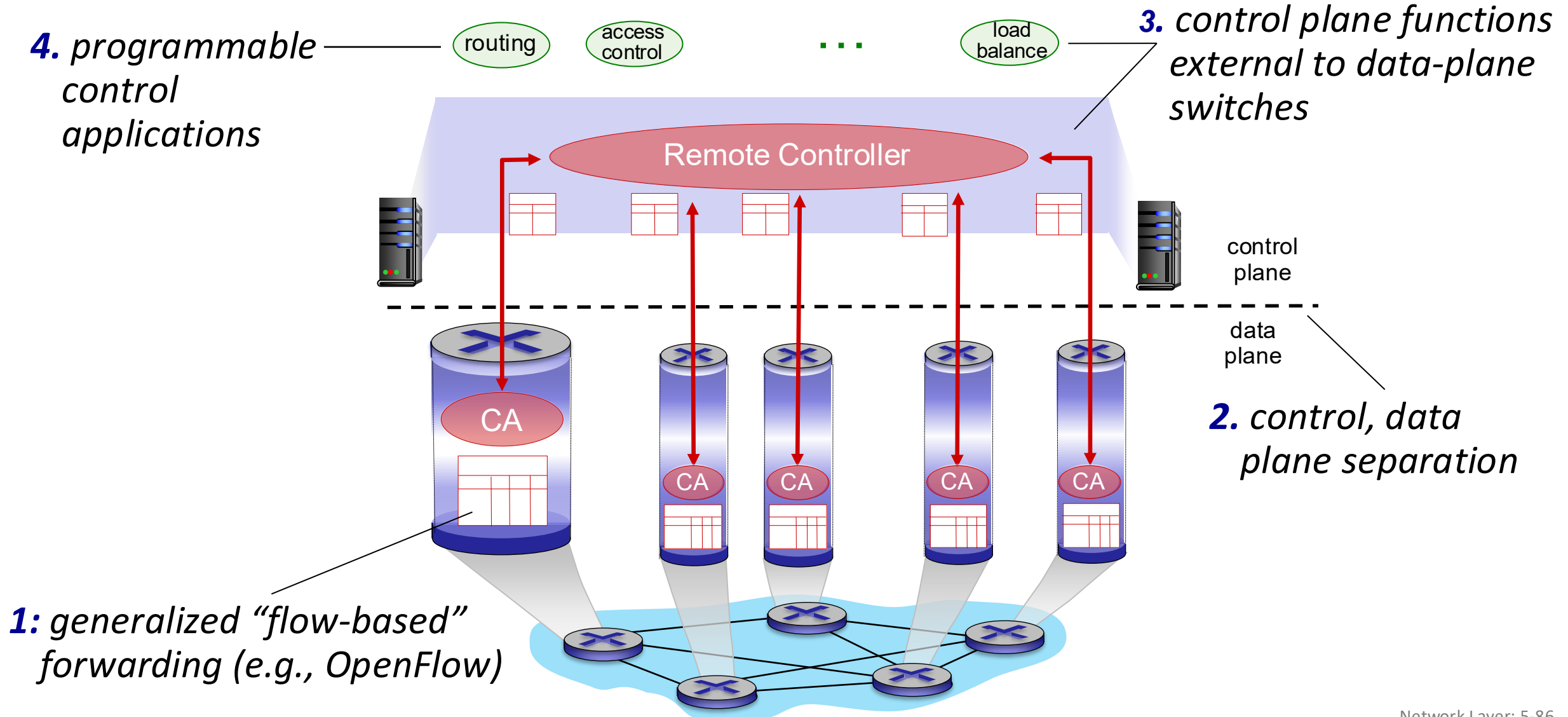


Q: what if w wants to route blue and red traffic differently from w to z?

A: can't do it (with destination-based forwarding, and LS, DV routing)

We learned in Chapter 4 that generalized forwarding and SDN can be used to achieve *any* routing desired

Software defined networking (SDN)



What is network management?

- autonomous systems (aka “network”): 1000s of interacting hardware/software components
- other complex systems requiring monitoring, configuration, control:
 - jet airplane, nuclear power plant, others?



"**Network management** includes the deployment, integration and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and element resources to meet the real-time, operational performance, and Quality of Service requirements at a reasonable cost."

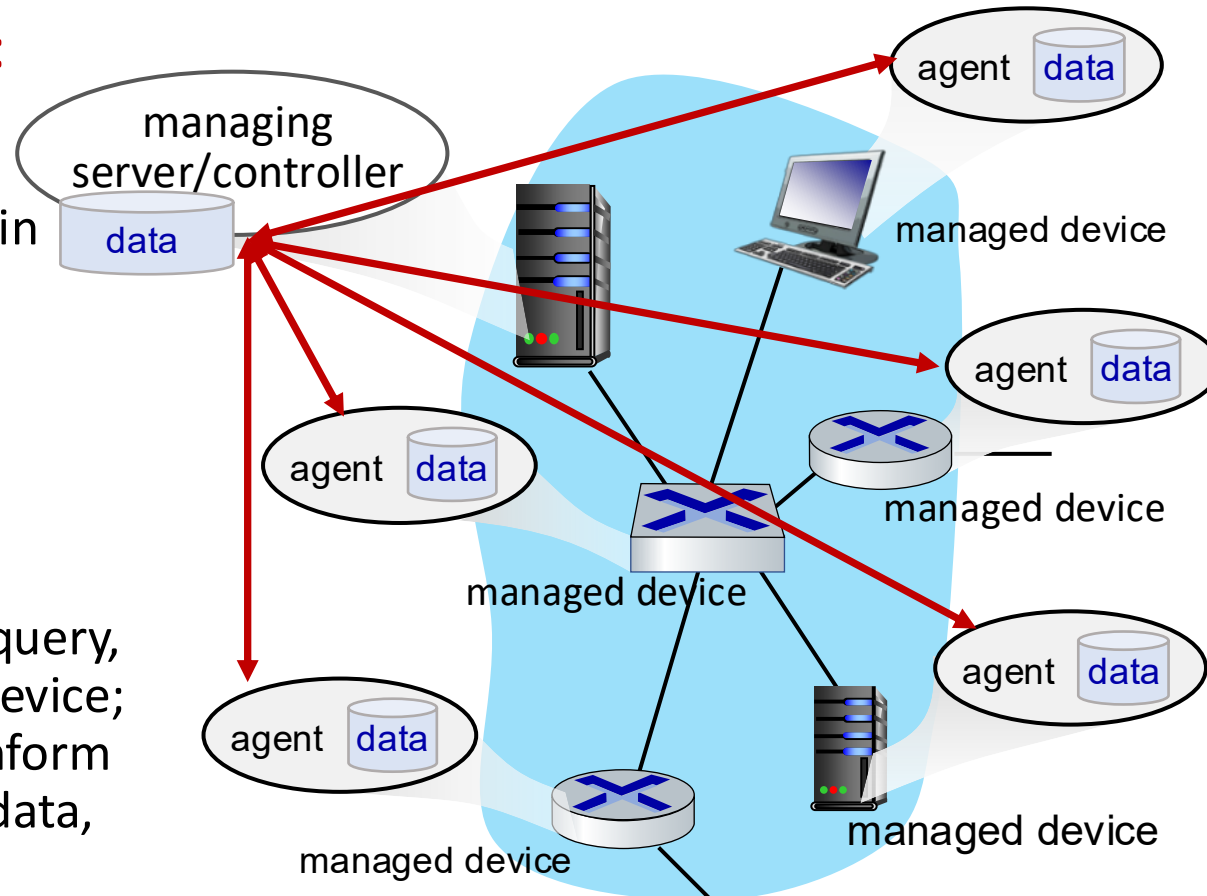
Components of network management

Managing server:

application, typically with network managers (humans) in the loop

Network management protocol:

used by managing server to query, configure, manage device; used by devices to inform managing server of data, events.



Managed device:

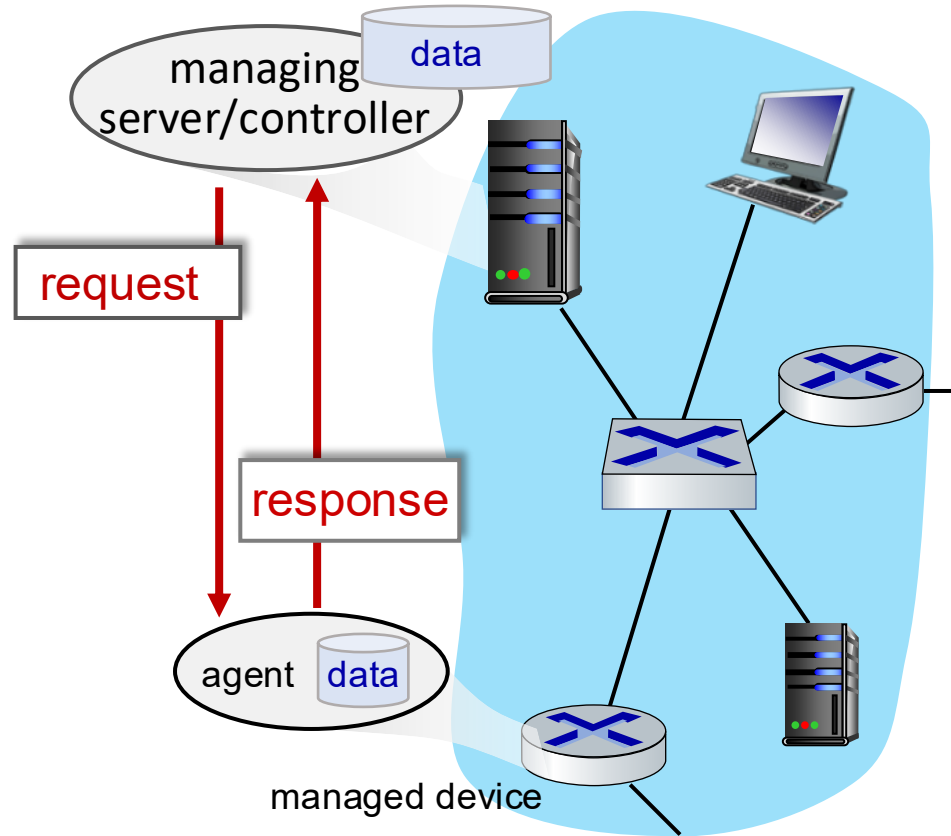
equipment with manageable, configurable hardware, software components

Data:

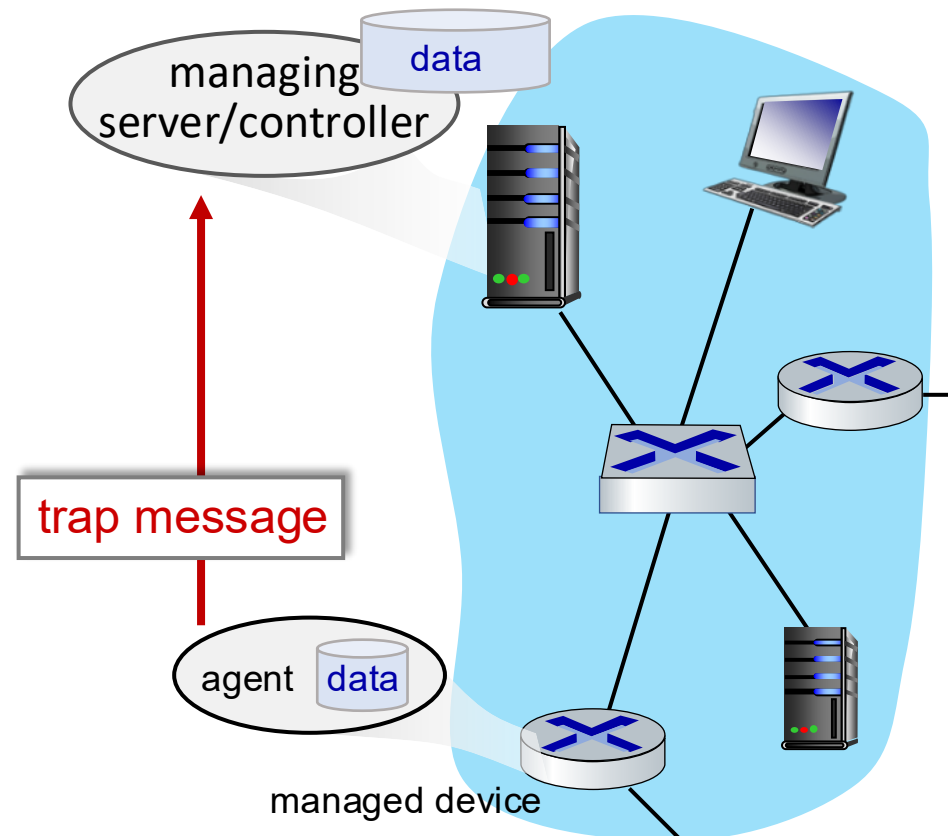
device “state” configuration data, operational data, device statistics

SNMP protocol

Two ways to convey MIB info, commands:



request/response mode



trap mode

SNMP: Management Information Base (MIB)

- managed device's operational (and some configuration) data
- gathered into device **MIB module**
 - 400 MIB modules defined in RFC's; many more vendor-specific MIBs
- **Structure of Management Information (SMI)**: data definition language
- example MIB variables for UDP protocol:



Object ID	Name	Type	Comments
1.3.6.1.2.1.7.1	UDPInDatagrams	32-bit counter	total # datagrams delivered
1.3.6.1.2.1.7.2	UDPNoPorts	32-bit counter	# undeliverable datagrams (no application at port)
1.3.6.1.2.1.7.3	UDInErrors	32-bit counter	# undeliverable datagrams (all other reasons)
1.3.6.1.2.1.7.4	UDPOutDatagrams	32-bit counter	total # datagrams sent
1.3.6.1.2.1.7.5	udpTable	SEQUENCE	one entry for each port currently in use